

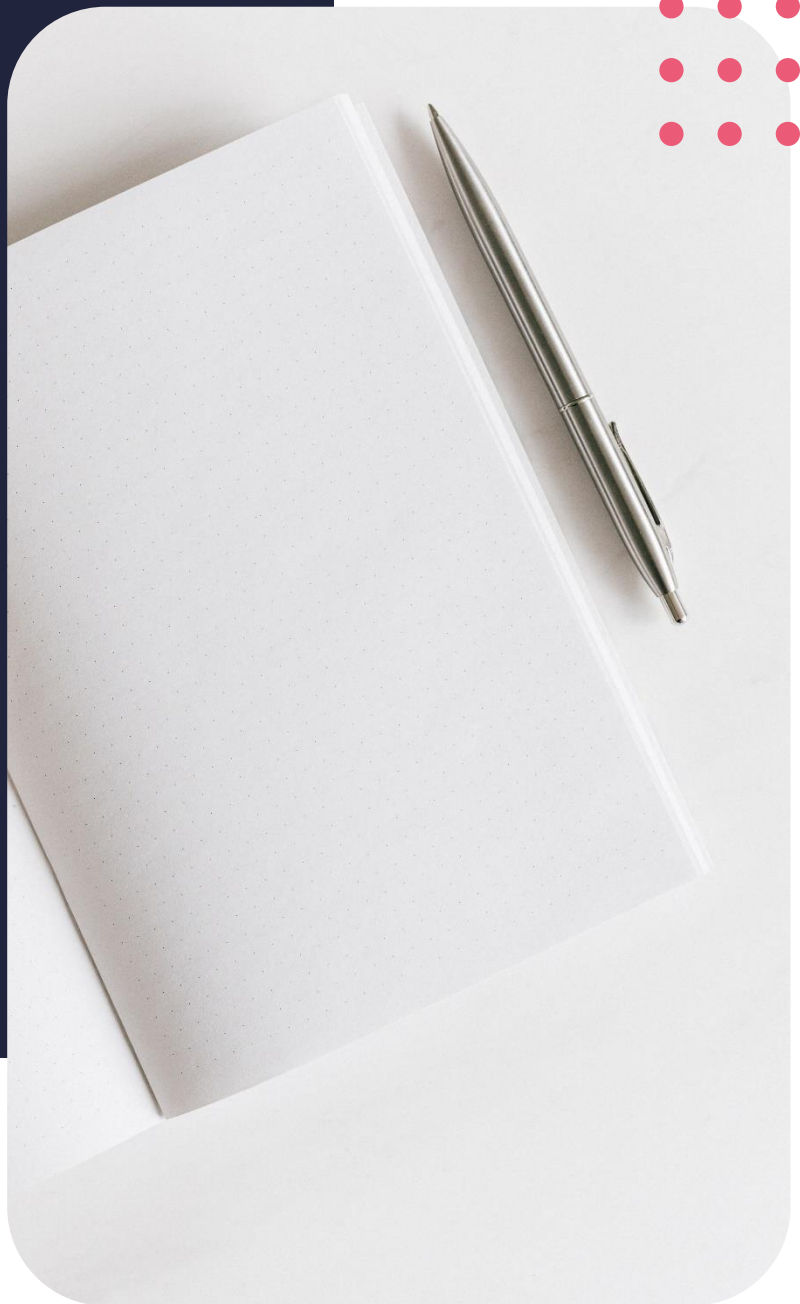
Diploma in Project
Management

Gantt Charts and Cost & Quality Management



Contents

3	Lesson objectives
3	Introduction
3	The Gannt chart
6	Cost management
9	Quality management
11	Conclusion



Lesson objectives

By the end of this lesson, you should be able to:

- A closer look at Gantt charts
- Cost management
- Cost estimating techniques
- Quality management

Introduction

A Gantt chart is a type of bar chart that represents your project schedule visually. The concept was first developed around 1910 by American Henry L. Gantt, who, working with a colleague Frederick Taylor devised a method of describing production planning and resource loading for factories and workshops.

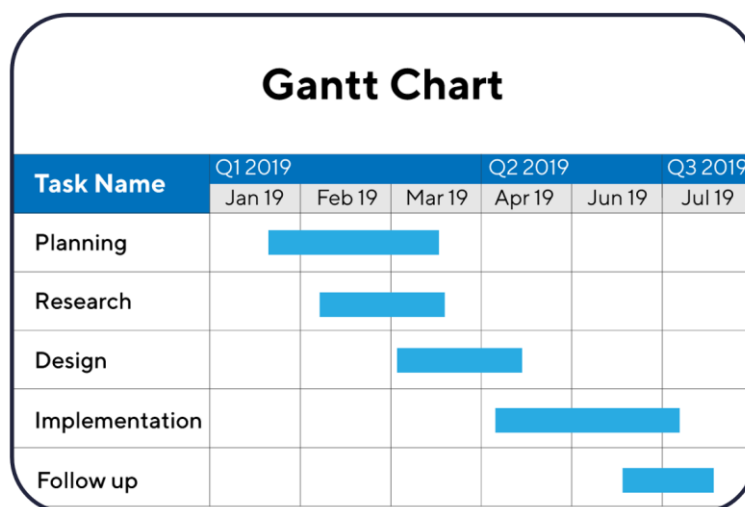
We start today's lesson with a closer look at the Gantt chart. We thought this would be useful following our discussion of planning and scheduling in the last lesson as it is one of the most commonly used tools for tracking project schedules today.

We will then talk about the importance of cost management and quality management in the project life cycle. And we'll also provide some more estimating techniques that may come in handy.

The Gantt chart

A Gantt chart is used for planning and scheduling of the project's tasks. As mentioned just now, it remains one of the most widely used project management tools – hence the need to cover it all on its own.

A Gantt chart is a type of bar chart that illustrates a project schedule.



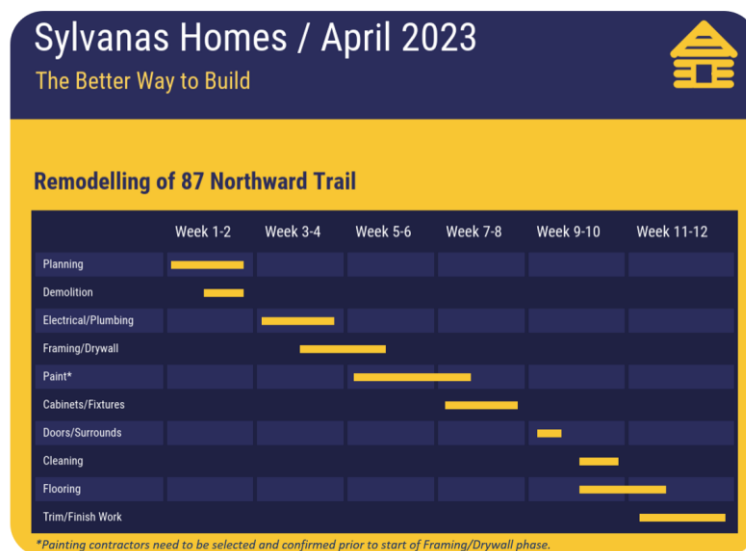
As can be seen in this very simplistic example, a Gantt chart lists the tasks to be performed on the vertical axis on the left, and start, end and duration as bars on the horizontal axis. The longer the bar, the longer the activity takes.

A Gant chart is also helpful for managing the dependencies between tasks. Do you recall our discussion about dependencies in the last lesson? Dependencies allow you to define relationships between tasks and to visualise the sequence in which they must be completed in order to close a project. You will remember that there are two steps to creating a dependency— you need a predecessor task and successor task and then you can establish a relationship between them.

Besides tracking dependencies some of the other benefits of the Gannt chart are that it:

- Improves team communication and productivity – because it gives live information on progress, the team can easily keep track of the project and will know what comes next.
- Helps with resource planning and allocation
- Summarises complex data to enable easy monitoring and tracking of progress
- Allows for project adjustments
- Identify risks and avoid problems

Let's look at a few examples of Gannt Charts...



This construction project Gannt chart example will make sure everyone's on the same page. The chart's simple format makes it easy to see *what* activity is due *when*.

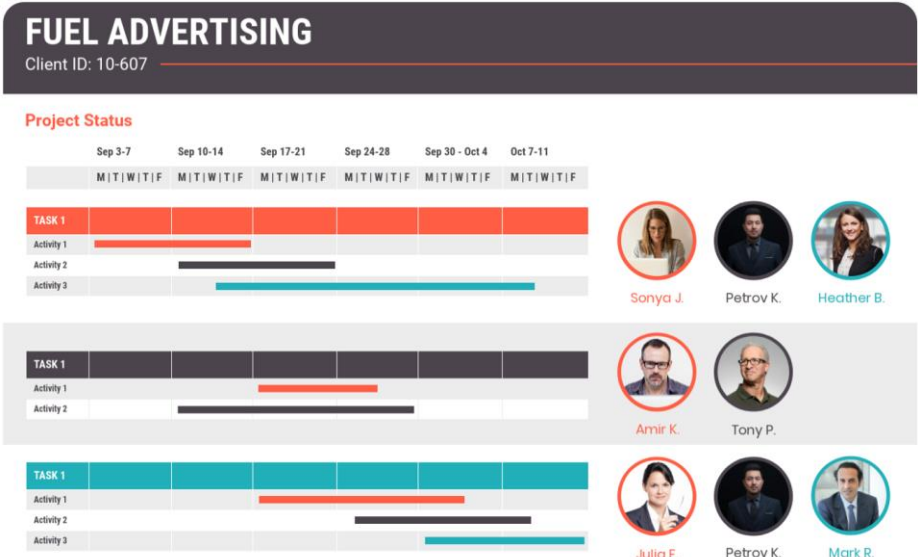
The beauty of using a good software package is that you can edit the chart as the project progresses: For example, if an activity is taking longer than it should, you could extend the bar's length and make that section red. That way, everyone is instantly aware of overtime and potential project delays.

A Gantt chart is the perfect way to manage tasks between multiple teams as all the information is in one place, in a crystal-clear format.



Take this example of a Gantt Chart for new product development. Product development typically involves multiple departments or teams of people. There may be no one project leader, only multiple managers on multiple teams. The far left hand column maps out the activities, while the far right column lists what department is responsible for each activity. A Gantt Chart in this scenario also eliminates confusion about dependencies i.e. who needs to finish what before another team can begin their task.

Gantt charts can be a helpful visual to include in project management plans, such as this. Here, the project activities are broken down day by day. t also breaks down *who's* responsible for *what* project activity.



- For example, Task 1/Activity 1 is color coded red.
- Sonya J. is responsible for it from September 3 to 14 (two weeks).
- Amir K. is responsible for it from September 17 to 21 (one week).
- Julia F. is in charge of Task 1/Activity 1 from September 17 to October 2. Her red bar shows that the due date is Wednesday.

Cost management

Cost management is the process estimating and controlling the different expenses within your project. It includes predicting future expenses in order to reduce the chances of going over budget. Project costs are calculated during the planning phase. Expenses are documented and tracked. Upon project completion predicted costs vs actual costs are compared.

There are four main phases of cost management: determining a budget, controlling costs, the different costs that are incurred during a project and estimating costs. Let's look at each phase in more detail.

Determine budget

The first step is determining your budget. So the first important thing to know is that project budgets are vital and allowing a project manager to make smart and informed decisions regarding scope, cost and time. The next important thing you need to know is that the budget takes into account all monetary resources needed to achieve your project goals. So your budget will include the different resources that you need, whether they be material objects, staff, resources, travel expenses. Anything incurred during the course of your project must be included in your budgets.

In order to establish a budget, you will need the following resources; You will need to come up with a risk register. This is something we will be covering in more detail during our lesson on risk. You will have a cost management plan. Your project schedule, which we discussed in the last lesson, your resource calendars also from lesson three, your scope baseline, so your expected scope throughout the project. And your cost estimates, which we are talking about today. When we say we need to control costs, what we mean by this is we need to monitor all costs in real time to ensure they stay aligned with the budget now in real time, what this means is exactly when it's happening, like in that moment, you need to make sure that your costs are running according to budget.

Control costs

When we say we need to control costs, what we mean by this is we need to monitor all costs in real time to ensure they stay aligned with the budget now in real time, what this means is exactly when it's happening, like in that moment, you need to make sure that your costs are running according to budget. It also means that you would

- Chart out task and team-based costs
- Track productivity to estimate billable hours
- Set task-based and hourly rates as required

The process of overseeing costs throughout the project allows opportunities to identify ways of reducing the expenses within the project which can in turn increase overall profits.

- It involves accounting for financial risks that could arise
 - If costs are not controlled, the project runs the risk of the project going over budget and not aligning with scope
-

Here's FIVE quick top tips for successful cost control:

- Capture the entire scope in your work breakdown structure. No task was too small or better yet, no work package is too small to include in your work breakdown structure. Have everything there and there are no surprises.
- Insist on inputs and collaboration from outside stakeholders. Remember what I keep telling you. All your stakeholders are very valuable resource within your project and they may have experience that differs from yours and therefore have insight that you don't yet have.
- Determine your different cost categories that are going to be used in the organisation. Cost categories or something will be covering in more detail later on in this lesson.
- But for now, just know, you need to determine your cost categories and before develop trust within your team. Inevitably, it will affect the productivity and the success of your project.
- Then you also need to take action immediately when there's a risk of exceeding your project budget. So the second you go over budget, you need to sound those alarm bells and figure out where all the money is going so that you can stop essentially the leak.

Different cost categories

We mentioned cost categories earlier, here is a brief summary of the different types of cost categories:

So if your project needs a new software, that would be a direct cost. So if you needed new software in order to build our e-commerce site, you would then have to go and purchase new software. Therefore you would have a direct cost. Then you've got indirect costs and these are not directly linked to your project. So these are things external to your project, but ultimately they will affect your project. So this could be, for instance, the rent of your office space, even though your particular project may not. You know, you may not consider the fact that you need a building to work and this is part of your indirect costs, something you do need. But that in the scope of your project is now then we've got fixed costs and these are ones of expenses. For example, you could buy a new company vehicle or buy a new computer or a new server. You are purchasing it once and then it's done. So one time cost is a fixed cost. Then you've got variable costs and these are subject to change suffixed, they stay variable, they can change. So this could be the gasoline prices that fluctuates, meaning that it's not a constant expense, that we've got sunk costs and these have already been paid. So, this could be doing the market research before launching a new product. Or in terms of your website, this could be the research done before taking on the clients. Any research you did into the company and their ability to pay for your services, you could then call a sunk cost and gathered a few key terms.

Some other 'cost' terminology that would be useful to know includes the following:

First, we have unburdened costs and these are the labour rates that are associated with money that you pay as a salary to your employees. So their fixed salary at the end of every month is called and unburdened cost. Then we've got burdened costs. And this is the cost of any additional benefits for your employees. So this could be health insurance, a car allowance, sulphur and any of that stuff that is added extras within your package is called a burden to cost. Then we've got material costs. This is your physical equipment that you are using at the office in order to do your job within your project, in order to do your job. So this could be your company, laptop service, any of that stuff would be considered as material cost. And lastly, we've got ancillary costs, which will be any other costs that are required, such as staff training. So if they need to learn a new programming language or learn how to use a computer, if they need to travel to go and meet the client and any other fees that they would be considered ancillary costs.

Estimating costs

The fourth phase of cost management is estimating. This is the process of forecasting what will be required in order to achieve your project scope. Remember that the forecast or estimate can often influence the decision to continue with the project or not. Remember the RAG review to determine if the item has a red, amber or green status?

It is important to note that your estimates do not need to be perfect. That is why they are called estimates. And also remember that your cost can be broken down into direct cost and indirect cost. Just a reminder; Direct costs are the costs linked directly to your project specifically. So this will be the software that you're using and your staff resources. Where are indirect costs? These are the costs that are incurred by the business itself? - renting your office space, having a receptionist there, your supplies in the kitchen.

Being able to accurately estimate the costs involved in a project is key to its success. We covered some estimating techniques in our last lesson but here we'd like to talk specifically about techniques that work particularly well when needing to estimate costs.

- Parametric estimation technique

Essentially, a **parametric estimate** is determined by identifying the unit cost or duration based on historical data and the number of units required for the project or activity. The main advantage of **parametric estimating** is that it is believed to have a higher accuracy than other types of **estimating** techniques and allowing **estimates** to be prepared in much less time. This estimation technique will most commonly be seen within construction work where you require materials and time that can be measured.

Let's look at an example: let's say your project requires a room that is 2.5 metres high and 10 metres wide to be painted. Historical data demonstrates that it takes 20 minutes to paint one square metre of wall and it uses one litre of paint.

So, what would our total square metre edge be? Well, we take that 2.5 metres high times the 10 metres wide and that all together will equal 25 square metres. Then what we need to do is go 25 square metres times 20 minutes is 500 minutes, but 500 minutes means nothing to anyone. So I've just divided that by 60. And that leaves us with eight hours and 33 minutes. Now, we've accounted for the labour. And now we need to work out how much paint we actually need.

You've got 25 square metres to paint and it takes one litre of paint per 25 metres. So in total, it's going to take us eight and a half hours or eight hours and 33 minutes and also 25 litres of paint, at which point we would then calculate the cost of eight and a half hours of labour and 25 litres of paint. So, there we have used the parametric example to estimate our costs.

- Analogous estimation technique

The analogous estimation technique uses a similar past project information to **estimate** the duration or cost of your current project, hence the word, "**analogy**". **Analogous estimation is an easy to use technique** when there is limited information regarding your current project.

A quick example to illustrate analogous estimation:

Tina's advertising agency is bidding on two different commercial campaigns. One is a digital campaign for a new resort in town, and the other is for a dating website.

The agency has done a digital advert campaign for a new hotel in the past, so they collect data from that campaign. The agency is better equipped to make an initial estimation for this project based on analogous estimating than the dating website campaign for which they do not have relevant past data available.

- Bell curve technique

The last estimating technique is the bell curve technique. Bell curves are used across a wide variety of disciplines, including finance and economics, social science, and the natural sciences. Analysts will often use bell curves when modeling different potential outcomes that are relevant for investing.

Just a quick recap on our bell curve technique and how it can be applied in terms of cost. We've got our X and Y axis here and our bell curve. Then remember, you've got your low, high and average. So, the lowest would be the cheapest that it can cost you or the least amount of money it can cost you. High would be the worst case scenario in terms of expense, and your average line in the middle would be the best-case scenario.

Quality management

You may be wondering why quality management features in project management. Quality management is the process for ensuring that all the activities necessary to design, plan and implement a project are effective and efficient.

When you're managing a project, the triple constraint is always at the top of your mind. But there is a fourth concern that is just as important, yet sometimes forgotten in the maelstrom of project management—*quality*.

It doesn't matter if you got the best software, met every milestone and completed the project under budget if the deliverable didn't meet quality standards. That's why project quality management is so important. What is quality? For project management, quality is simply what the customer or stakeholder needs from the project deliverables. By keeping the definition tied to the customer or stakeholder, quality management can have a narrower focus, which means it's more likely to achieve its goals.

So, what would be the key focus areas of quality management? The first one that springs to my mind is customer satisfaction. Now, in the planning phase, you need to ensure that you clearly understand what your customer requires. And then remember that by finding out the requirements and ensuring that they're clear, you can then deliver the end result that satisfies everyone involved in the project by having your requirements clearly defined. It also really helps to build a positive relationship with your clients. You always want to leave your customer happy because word of mouth is the best way of getting more work in.

Did you know?

You probably have heard many different figures bandied about with this question, but what percentage of customers do you think actually complain when they are unhappy about a product or service?

According to the White House Office of Consumer Affairs, for every customer who bothers to complain, 26 customers remain silent. That means that only 4 percent of dissatisfied customers complain, while 96 percent will never let you know that there was a problem.

Tips for satisfied customers

- Always keep your customer informed, make sure they know what's going on. There is nothing worse than asking someone to do something for you and you have no idea where they are, if they're even doing it. You can do this by providing progress checks. You can even have them in for weekly meetings where you demonstrate your improvements or your progression and then always make sure that you return phone calls or emails with your customer.
- Always under promise and over deliver. You'd rather pleasantly surprised someone than disappoint them. It is also vital that you take ownership even when things are going wrong. So don't only boast about your successes within the project, but if something goes wrong or if an element isn't working, take ownership as the project manager. If things are going wrong, it is on you! Make sure that you don't disregard what the customers are saying because you know better. They're the ones paying for it. If you can suggest try to advise, but don't bash your head against a wall trying to get them to understand you if they want something done a certain way.

Quality Management Plan (QMP)

One good way to keep control of all of this is to have a quality plan, so this is going to allow you as a project manager to track the following elements. Firstly, are you aligning with the customer's requirements? Are you delivering the things that you have now promised to deliver and that your customer needs? Are you aligning with your internal and external stakeholders? In other words, do they know what's going on and agree with the methods you are using in order to reach your project goals? And do they have any suggestions that you could take on board your members? Satisfied stakeholders are very important to your project success. Then you also need to track problems within the project. Now, what's important here with a quality plan is to also be able to identify when exactly the problem started. So rather than at the end

of the project to see something went wrong, you want to know straight away. Here is where things went wrong. So here's where our quality got worse, the overall performance of the project plan.

Now, the quality management plan usually includes the following quality management components. What are your objectives within your quality? So what do you aim to achieve in terms of quality, your key project deliverables and processes that are going to be reviewed for your quality level later? So what things are you actually delivering in? Is it going to satisfy the customer's requirements? You need to set your own standard as well as your client's standard for the quality that you deliver.] And then you need to have quality control and assurance activities. Quality roles and responsibilities also need to be considered, as well as the different tools that you can use in order to manage the quality of your project. And then you also want to plan for reporting quality control and assurance problems.

So, basically, the purpose of the QMP is to describe how quality will be managed throughout the lifecycle of the project. Quality management planning determines quality policies and procedures relevant to the project for both project deliverables and project processes, defines who is responsible for what, and documents compliance.

Quality assurance

We mentioned quality control and assurance. Let's look at those two concepts quickly.

What is quality assurance? Well, it's process orientated. It focuses on ensuring that the desired quality requirements are met. And how we do this is by evaluating different processes. So the steps in which you need to complete tasks and activities and also procedures, which is a defined set of activities when certain events occur.. And this process is also going to help prevent defects or deviations from your requirements, so by evaluating where you are and the quality of your project, you can see if there are any problems arising at this point.

Quality control

Now, how does this quality assurance differ from quality control? Quality control is more product oriented. And it focuses directly on the product or deliverable in relations to the specifications at different stages of the project. So take our website building example we used in a previous lesson. We are going to focus on the website and it's going to compare to any specifications made within your project. So the different deliverables and milestones, the new projects and measures against that SEO per week, three, we should have the home page done and then you would go and evaluate that home page in comparison to the requirements that you've been given.

This process must be present throughout the life cycle of the project so that adjustments can be made. So you want to identify the problem soon so that you can fix them as soon as possible, so that you can lessen the impact on the rest of your project. As I said earlier, you want to do this during the project rather than at the end, because by then one tiny bug can grow into this awful big problem within your project. So rather handle things as they go and spot quality. That is a problem early on so that you can fix it before to fix the rest of your project.

User acceptance testing

This form of testing you'll find more in the IT industry when you're testing software or websites or basically anything Artie related. And the point of this whole process is that designated testers are challenged with breaking or finding gaps in your system's functionality or design. So you finish something and you're essentially asking them to go and make sure that your code, whatever you've done, is rock solid and that it can't be broken through this process. They'll pick up bugs or defects and they are captured. And then developers and designers need to go and then find a solution to the potential problem or the problem that is occurring already in your programme.

The solution they come up with is then tested again until it meets user requirements. Now, this will sound a bit familiar in terms of how we discussed agile development within the first lesson. So it's the process of testing and then trying to fix any problems until eventually your test goes completely smoothly, perfectly, and your product gets signed off.

Conclusion

Before we go, let's recap what we have done and look at what we will cover in upcoming lessons.

In our first lesson, we introduced you to the project management cycle.

The second lesson focused on the Initiation Phase of the lifecycle.

Then we moved onto the Project Plan. This lesson on Cost and Quality Management would fall under the planning phase of the project.

As will the next lesson on Control, Risk, Procurement and Acceptance Management.

Then we move onto the next phase of the lifecycle – project execution and here we will touch on leadership. Monitoring and controlling is the next phase and the final stop will be the project closure phase of the life cycle.
